Selected versus skipped curves: two open hypotheses

21 October 2026. Exploratory six-curve diagnostic and prospective research plan.

The questions are (H1) whether the published curve has an exploitable weakness, and (H2) whether an exploitable weakness in earlier eligible curves explains why they were skipped. Both hypotheses remain open. They are not mutually exclusive, and an undocumented engineering rule is a separate explanation.

Scope and identities

The prior census and its exact order rows identify these within-stream cohorts:

• P-224, `srand48(891607927)`: eligible indices 777, 1485, 5736 and 7026; index 7026 is the published curve.

• P-256, `srand48(871696354)`: eligible indices 876 and 1842; index 1842 is the published curve.

Here **eligible** means passing the recorded known checks: a nonsingular seed-derived model, prime full order, permitted order bit length, non-anomalous order and no embedding degree below 1024. It is not an attestation of the historical decision process. **Skipped** means an earlier eligible candidate that was not published. The many blocks failing the known checks form a separate population; finding composite orders among those blocks would not explain skipping these four eligible candidates.

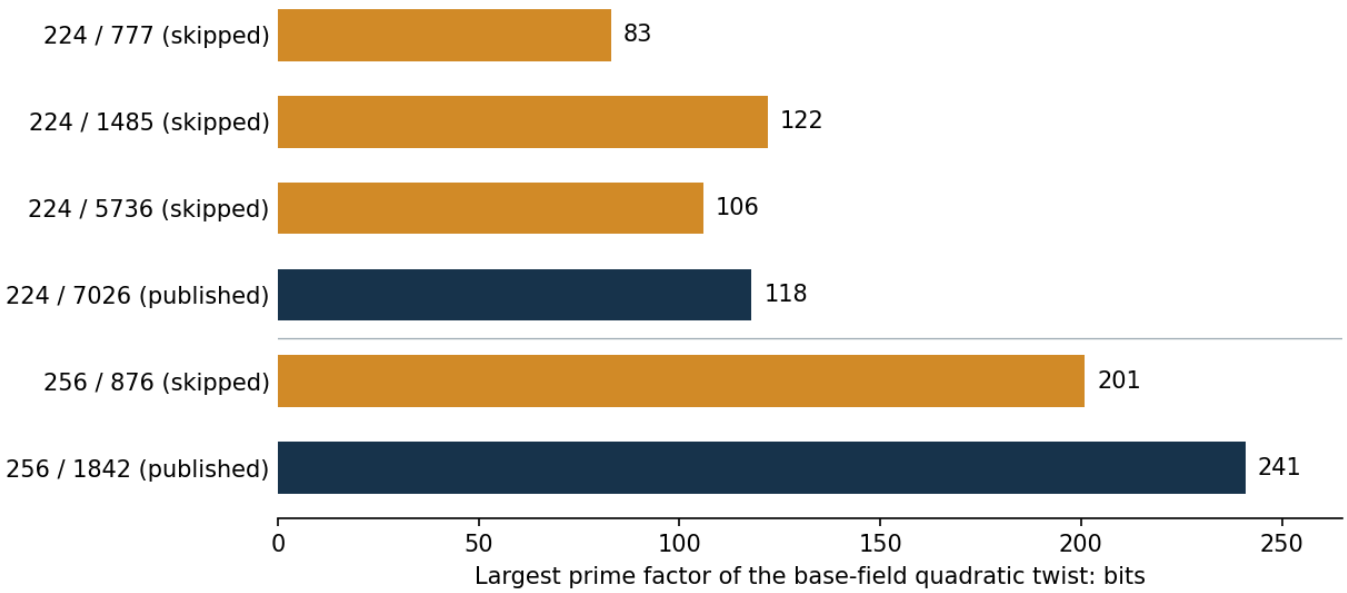
inputs.tsv freezes the exact models, supplied or standardized generators, subgroup orders and full ICV1 identities. sources.json pins the census, standard parameter and identity sources to a commit and Git blob IDs. Standard generators are retained; their unexplained historical selection is not replaced with the first-x rule used for the supplied curves.

Observed arithmetic comparison

The native C++ diagnostic uses PARI's C library and OpenSSL, with no Python research execution. Each of the six rows proves field and order primality, reconstructs its stream seed and X9.62 relation, checks the generator and unique multiple of its order in the Hasse interval, and checks that the embedding degree is at least 1024. Complete factorizations below are checked by prime proofs and exact products. No DLP recovery or timing comparison ran.

Field / index	Role	Largest twist prime: bits	Twist work: log2, approximate	Largest prime in n-1: bits	Largest prime in n+1: bits	Frobenius-order conductor
224 / 777	skipped	83	41.50	145	182	1
224 / 1485	skipped	122	60.54	74	102	1
224 / 5736	skipped	106	52.69	204	142	1
224 / 7026	published	118	58.55	196	177	3
256 / 876	skipped	201	100.15	242	232	1
256 / 1842	published	241	120.45	92	215	1

The twist work column is half the logarithm of the largest twist prime; it estimates generic group operations, without a measured attack or a protocol acceptance assumption. Main-curve square-root estimates remain about 112 and 128 bits. Records are in records/.



What these differences discriminate

Twists. P-224's published curve has a smaller largest twist prime than skipped index 1485. Thus the rule "publish the eligible candidate with the largest twist prime" does not explain the P-224 choice. P-256's published curve has the larger twist prime of its pair. These facts do not exclude a threshold combined with other filters, or a weakness in another mechanism. Twist exposure requires a protocol or implementation accepting twist points; it is a different hypothesis from solving the valid prime-order curve's DLP.

Nearby integers. The published P-256 order has a much smoother $n-1$ than skipped index 876 (largest factors 92 versus 242 bits). Conversely, P-224 index 1485 has the smoothest $n-1$ and $n+1$ in its cohort. A Cheon-style route needs the specified auxiliary powers or an applicable oracle. The usual pair $(G, [x]G)$ alone does not supply that input. Smoothness of $n-1$ is not a Pohlig-Hellman factorization of the prime group order n.

P-224 structure. The published curve's Frobenius discriminant has square part 9, giving Frobenius-order conductor 3. All three skipped P-224 curves have fundamental Frobenius discriminants and conductor 1. The published curve's 3-division polynomial splits into four linear factors; each skipped curve has factor degrees 1 and 3. The native records also retain the square characters of the corresponding rational x-coordinate right-hand sides. For the published curve, all four characters are -1: the full nonzero 3-torsion has coordinates over the quadratic extension, not the base field. These are concrete inputs for a local 3-isogeny/endomorphism investigation. Conductor 3 here refers to the order generated by Frobenius; it must not be silently equated with the actual endomorphism-ring conductor. No executable attack follows from the difference or from extension-field small torsion alone.

Selection direction. A single rule "the published curve is stronger on every listed proxy" does not fit these observations. Neither does a single rule "the published curve is weaker on every proxy." This six-curve, unblinded comparison reports arithmetic differences, not a statistical outlier finding, a historical motive or evidence of a trapdoor.

Separate obligations for H1 and H2

Track	Evidence that would advance it	Current status
H1: published-curve ordinary-input DLP weakness	An explicit applicable reduction or algorithm, preserved prime subgroup, verified recovery and complete cost against matched rho	Not established; routine checks pass
H1: retained secret witness or hidden route	Specify witness, public reconstruction alternatives, map construction/evaluation, subgroup action, destination solver and recovery	No such witness or route supplied or constructed
H1: conditional protocol exposure	A protocol that actually supplies the twist or auxiliary-power input and an applicable attack with full cost	Preconditions untested

Track	Evidence that would advance it	Current status
H2: weakness of earlier eligible curves	The same attack contract tested on every eligible candidate and the published control in each stream	Four earlier curves checked arithmetically; no ordinary-input attack established
H2: weakness explains rejection	A rule frozen before new results, explaining the skipped labels and predicting held-out choices beyond known filters	Unexplained; prior census's exploratory skip-filter tests found no separating rule

A meaningful advantage may remain expensive: an intermediate exponent below one half is within scope. Separate algorithm existence, executable transfer, correctness, public discovery cost, retained-witness setup, per-target work, reuse, memory and hardware. An efficient known-scalar multiplication or an available small isogeny does not itself establish unknown-scalar recovery. An unpublished family-wide method could apply to both selected and skipped curves; absence of a selection anomaly would not exclude that scenario.

Recovering the drand48 seed stream narrows a generation model. It neither proves nor disproves a curve-specific weakness, a hidden family-wide method or a private construction witness. Predictable public parameter generation does not imply predictable private keys or signing nonces.

Next discriminating work

1. **Complete the matched trait baseline.** Port the corrected trait path to native code and apply the same 22-trait parameter grid to both published curves and the supplied four, preserving partial results and the original frozen outputs. This six-curve diagnostic is a subset, not that full run.

2. **Probe the P-224 structural difference.** Use the four rational 3-division x roots, their y characters and the certified discriminant to determine the actual local endomorphism order. For a proposed route, retain kernels, explicit maps, their working fields and action on the original prime subgroup. Test a map/dual composition as a scalar control. A faster unknown-scalar route needs its own reduction and complete cost evidence.

3. **Make nearby-order leads conditional and executable.** For each factor choice in `n-1` or `n+1`, specify the exact auxiliary points and queries required. Establish whether the proposed protocol supplies them before interpreting any attack complexity. Validate applicable algorithms on public known-answer fixtures; keep full-size extrapolation separate.

4. **Broaden the selection test.** The pinned census has 21 eligible prime curves: 10 published and 11 skipped. These six cover two published and four skipped curves; profile the other 15 with the identical native feature definitions. Treat that census as exploratory because its outcomes and earlier feature searches are already public. Commit hypotheses, primary scores, control sampling and a multiple-testing policy before generating fresh matched streams for a prospective test. Condition on field, `a=-3`, root convention and every known acceptance rule; do not pool raw bit sizes.

The exact unknown selection rule and missing attack witnesses are tracked in the problem record. These next steps are planned, not background jobs or completed experiments.

Validation, limits and reproduction

```
PARI_PREFIX=/path/to/pari sh native/run.sh
sh native/controls.sh
```

The driver builds the C++ code, runs each of six fixed rows with a 90-second process limit, and stops on failure. Inputs are the exact frozen table. No factorization residual is accepted as prime: every factor is proved and its product checked. Native seed checks include adjacent-run and adjacent-block negative controls. Additional fixed seed, coefficient, point and order corruptions must fail without producing a result. The initial native API integration failure and its correction are preserved under `records/`.

The original four-curve Sage evidence is unchanged. This follow-up's native computations are one-host checks using a shared PARI backend, not an independent SEA implementation, independent-host replay or Lean proof. Cargo and Conductor are unavailable in this runtime; attempted commands and their actual outputs are retained. This limits workflow verification, not the definition of either hypothesis.

The figure plots recorded integer bit lengths for six curves from two streams, with no statistical interval inferred. Plotting and PDF code only renders existing records. The original assessment graph and PDF, active curve catalogue, scoreboard and leaderboard remain unchanged: this adds a diagnostic supplement, with no solver timing or performance promotion.

Exact model identities

P224-candidate-777 · skipped


```
ICV1:fp-269599466671506397946670150870196306735579162600263081435100662988813901553279810317031433354100
437823:269599466671506397946670150870196267720046364497092767101559658610592258248209711262442372527302505
052313056593101247937126324318365272245:-926176686734039171949747491355849919944237301968119317895999762116
6195:unk:r:4b5d3f89ec3f
```


P224-candidate-1485 · skipped


```
ICV1:fp-269599466671506397946670150870196306735579162600263081435100662988813601449797251772972095796455
03161:269599466671506397946670150870196305134129365348490109339304207957212414094358970078725150581636901925489261505406038054085015433404783 2:-107814140254071381731076368453834469357966102414424367941271404203603:unk:r:eb89557e65b2
```


P224-candidate-5736 · skipped


```
ICV1:fp-269599466671506397946670150870196306735579162600263081435100662988813224555599954095973222272472
665529:269599466671506397946670150870196274490023163059303349212375936333531080619981379884906279159097913847866111043073446213087756184582252 9:-9744202785140723935188329567548515544053065039837095023756196034
5683:unk:r:b792075970b9
```


P-224-published · published


```
ICV1:fp-269599466671506397946670150870196306735579162600263081435100662988814733100108545601916421827343
930821:269599466671506397946670150870196259404578077144243917216827223680612078197707962899647706300737973484905751973224228719493668660579467 7:unk:unk:r:fec01e99874c
```


P256-candidate-876 · skipped


```
ICV1:fp-115792089210356248762697446949407573530086143415290314195533631308867097853951:105867235723906477
692553525193676385009:115792089210356248762697446949407573529980276179566407717841077783673421468943:1666
431558791302003980351035011959112391866348083660076214606213177407344151 1:-45196048524160381514294955574887608279554759907305149886414376967531399148572 3:unk:r:a67f0bc0376d
```


P-256-published · published


```
ICV1:fp-115792089210356248762697446949407573530086143415290314195533631308867097853951:891881911545538531
113722477985858095 83:115792089210356248762697446949407573529996955224135760342422259061068512044369:79589
093771320884530747432173573986150410652824946103043721159066269675301 47:unk:unk:r:f188c4914bb7
```


Exact model, generator and subgroup parameters are frozen in inputs.tsv; source and output hashes accompany this supplement.